

Technical Note: Sentiment-Driven Portfolio Strategy

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Introduction

This technical note documents the development, implementation, and evaluation of a sentiment-driven portfolio strategy for the South African equity market. The strategy uses the Kasi Composite Confidence Index (CCI) as a macro-sentiment signal and combines it with JSE sector indices to dynamically allocate capital across market regimes.

The primary objective is to evaluate whether incorporating consumer sentiment data can improve portfolio risk-adjusted performance relative to a passive benchmark. The methodology applies regime classification based on historical CCI percentiles, sector allocation informed by recent momentum, and a comprehensive set of risk and performance metrics. Strategy performance is evaluated against the ALSI Total Return index over the available sample period.

This document is intended for quantitative researchers, portfolio managers, and practitioners interested in the application of sentiment indicators to systematic asset allocation in emerging markets. The analysis was conducted using reproducible code and data workflows; further details and materials are available upon request.

Data & Methodology

Data Sources

The analysis utilizes three primary data sources:

1. **Kasi Composite Confidence Index (CCI):** A proprietary macroeconomic sentiment indicator reflecting consumer and business confidence in South Africa. The CCI is published monthly and serves as the basis for regime classification.
2. **JSE Sector Indices:** Total return indices for major sectors of the Johannesburg Stock Exchange, including Financials, Industrials, Resources, and others. These indices provide the universe for sector allocation and momentum analysis.
3. **ALSI Total Return Index:** The FTSE/JSE All Share Index (ALSI) total return series is used as the passive benchmark for performance comparison.

Data Preprocessing

All datasets were imported in monthly frequency and aligned on calendar month-end dates. Preprocessing steps included:

- **Index Exclusion:** The SWIX index was excluded from analysis due to limited data availability (only from December 2020 onward).
- **Data Cleaning:** Removal of duplicate entries, standardization of index names, and verification of time series continuity.
- **Date Adjustments:** The analysis period begins February 2016, with the first month (January 2016) dropped due to NaN values from initial return calculations. All indices and sentiment data were merged on nearest month-end dates to ensure consistent alignment.
- **Missing Data Handling:** No forward-filling was applied; periods or indices with insufficient data were excluded from the final dataset.

Regime Classification

Market regimes are assigned monthly based on the percentile rank of the Kasi Composite Confidence Index (CCI) within the historical sample. The classification logic is as follows:

- **RISK-ON:** CCI value is above the 67th percentile (top third of historical values).
- **NEUTRAL:** CCI value is between the 33rd and 67th percentiles.
- **RISK-OFF:** CCI value is below the 33rd percentile (bottom third of historical values).

This approach allows the strategy to adapt dynamically to shifts in macro sentiment, with regime boundaries chosen to ensure a balanced distribution of market states over time. The percentile-based method is robust to outliers and changes in the absolute level of the CCI.

The assigned regime for each month determines the eligible sector indices for allocation and triggers portfolio rebalancing only when a regime change occurs.

Sector Allocation & Momentum

Within each regime, sector allocation is determined using a momentum-based approach:

1. **Eligible Sectors:** For each regime (RISK-ON, NEUTRAL, RISK-OFF), a predefined set of JSE sector indices is considered based on their historical performance in different macro environments.

2. **Momentum Calculation:**

- For each eligible sector i , 6-month total return momentum is calculated as:

$$extMomentum_i = \frac{P_t}{P_{t-6}} - 1$$

where P_t is the sector index level at month t and P_{t-6} is the level six months prior.

- **Example:** If Industrials index was 10,000 six months ago and is 11,200 now, then:

$$extMomentum_{\text{Industrials}} = \frac{11,200}{10,000} - 1 = 0.12 \text{ (12\%)}$$

3. **Selection & Weighting:**

- Sectors with positive momentum are included in the allocation; those with negative or missing momentum receive zero weight.
- If all eligible sectors have negative or missing momentum, allocation is split equally among them.
- Otherwise, weights are assigned in proportion to positive momentum values:

$$w_i = \frac{\text{Momentum}_i}{\sum_{j \in S^+} \text{Momentum}_j}$$

where S^+ is the set of eligible sectors with positive momentum.

- **Example:** Suppose in a RISK-ON regime, the eligible sectors and their 6-month momentum are:
 - Top 40: 8%
 - Telcos: 3%
 - Industrials: -2%
 - Discretionary: 1% Only Top 40, Telcos, and Discretionary are included. Their weights are:

$$w_{\text{Top 40}} = \frac{0.08}{0.08 + 0.03 + 0.01} = 0.666 \text{ (66.6\%)}$$

$$w_{\text{Telcos}} = \frac{0.03}{0.08 + 0.03 + 0.01} = 0.25 \text{ (25\%)}$$

$$w_{\text{Discretionary}} = \frac{0.01}{0.08 + 0.03 + 0.01} = 0.083 \text{ (8.3\%)}$$

4. **Rebalancing:** Portfolio weights are updated only when a regime change occurs, not on every month.

Risk and Performance Metrics

The strategy is evaluated using a comprehensive set of risk and performance metrics, designed to capture both absolute and risk-adjusted returns, as well as downside risk and market sensitivity. The key metrics include:

Annualized Return

The annualized return measures the geometric average return earned by the strategy each year over the evaluation period. It is calculated as:

$$\text{Annualized Return} = \left(\prod_{t=1}^T (1 + r_t) \right)^{\frac{12}{T}} - 1$$

where r_t is the monthly return and T is the total number of months.

Cumulative Return

Cumulative return represents the total percentage gain or loss over the entire evaluation period:

$$\text{Cumulative Return} = \prod_{t=1}^T (1 + r_t) - 1$$

where r_t is the monthly return.

Volatility (Standard Deviation)

Volatility measures the dispersion of returns and is a proxy for the risk of the strategy. It is calculated as the annualized standard deviation of monthly returns:

$$\text{Volatility} = \sigma_{\text{monthly}} \times \sqrt{12}$$

where σ_{monthly} is the standard deviation of monthly returns.

Sharpe Ratio

The Sharpe ratio measures the risk-adjusted return of the strategy, defined as the excess return per unit of volatility. It is calculated as:

$$\text{Sharpe Ratio} = \frac{\bar{r} - r_f}{\sigma}$$

where $\bar{r}$ is the average monthly return (annualized), r_f is the risk-free rate (annualized, typically set to zero for simplicity), and σ is the annualized volatility.

Up/Down Capture Ratios

Up and Down Capture Ratios measure the strategy's performance relative to the benchmark during periods when the benchmark is rising (up) or falling (down).

- **Up Capture Ratio:**

$$\text{Up Capture} = \frac{\sum_{t \in U} r_{\text{strategy},t}}{\sum_{t \in U} r_{\text{benchmark},t}}$$

where U is the set of months when the benchmark return is positive.

- **Down Capture Ratio:**

$$\text{Down Capture} = \frac{\sum_{t \in D} r_{\text{strategy},t}}{\sum_{t \in D} r_{\text{benchmark},t}}$$

where D is the set of months when the benchmark return is negative.

A ratio below 100% in down markets and above 100% in up markets is desirable.

Maximum Drawdown

Maximum drawdown measures the largest peak-to-trough decline in the portfolio value over the evaluation period. It quantifies the worst observed loss from a historical high:

$$\text{Max Drawdown} = \min_t \left(\frac{V_t - \max_{s \leq t} V_s}{\max_{s \leq t} V_s} \right)$$

where V_t is the portfolio value at time t .

A lower maximum drawdown indicates better downside risk protection.

Results

Performance Analysis

This section evaluates the sentiment-driven strategy's performance versus the ALSI benchmark using a suite of professional analytics and visualizations. Each result is accompanied by a concise, informative interpretation to guide investment insight.

Cumulative Returns

The chart below compares the cumulative returns of the sentiment-driven strategy and the ALSI benchmark over the backtest period.

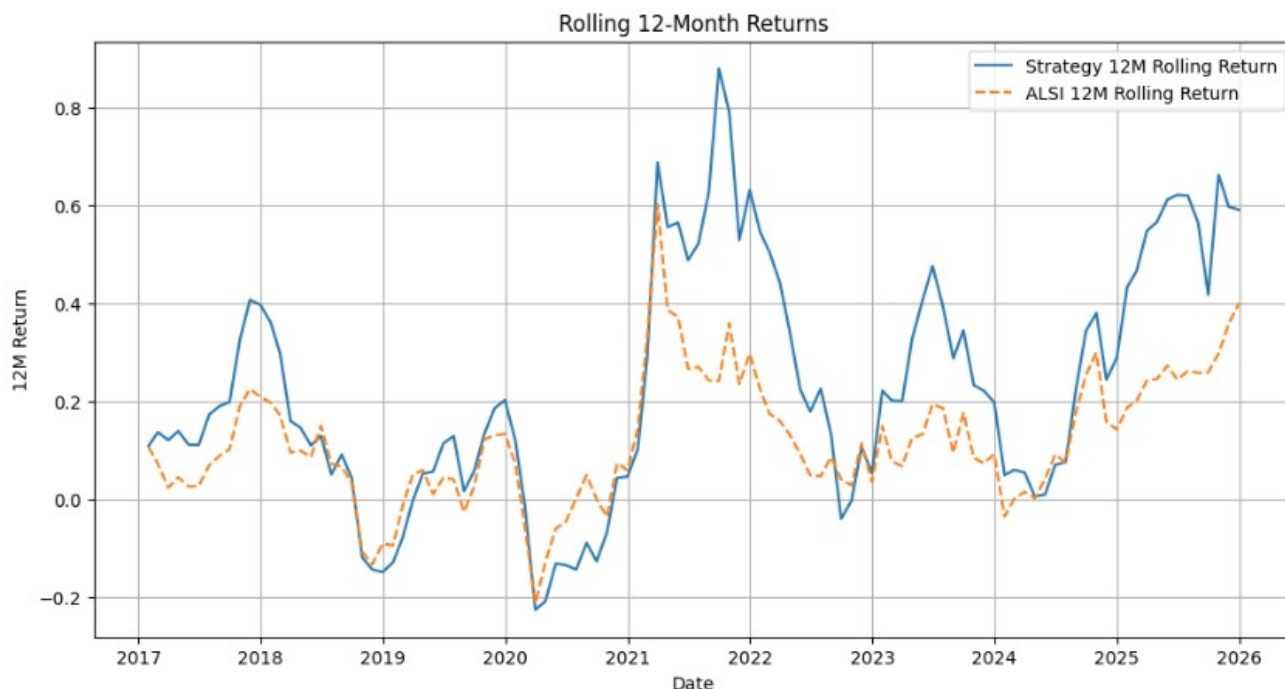


Interpretation: The cumulative return profile indicates that the sentiment-driven strategy delivers persistent and economically meaningful outperformance relative to the ALSI benchmark over the full backtest horizon. The divergence between the two curves becomes particularly pronounced following major market stress events, suggesting that regime-based reallocation and momentum filtering contribute to improved capital deployment during recovery phases.

Periods of relative underperformance or convergence are generally short-lived and tend to coincide with transitional macro environments, where sentiment signals adjust more gradually and sector leadership is less clearly defined. Importantly, the strategy avoids prolonged drawdowns relative to the benchmark, indicating that downside risk is effectively mitigated through defensive positioning during adverse sentiment regimes.

Rolling 12-Month Returns

Rolling 12-month (1-year) returns reveal how performance evolves through time, smoothing short-term noise.

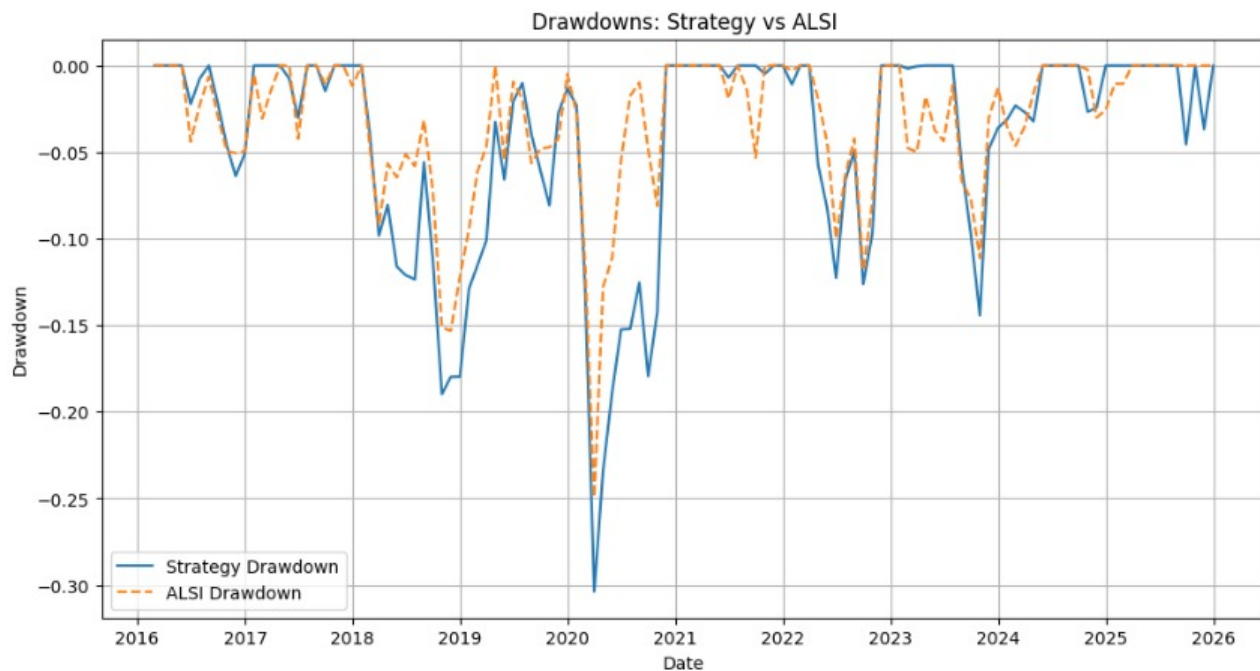


Interpretation: The rolling 12-month return comparison shows that the sentiment-driven strategy consistently achieves higher return amplitudes than the ALSI benchmark across multiple market cycles. Periods of pronounced outperformance most notably post-2020 and again from 2023 onward suggest that the strategy is able to scale exposure effectively during favorable macro-sentiment regimes, translating into stronger medium-term performance than a passive allocation.

Importantly, the strategy's rolling returns do not merely track the benchmark at a higher level; they also exhibit earlier recoveries following market drawdowns, indicating improved responsiveness to changing sentiment conditions. While underperformance is observed during certain transition periods (e.g., around regime inflection points), these phases tend to be shorter-lived and less severe than the benchmark's downturns. The rolling return profile shows that excess performance tends to persist once established, with periods of outperformance clustering over time rather than appearing as short-lived spikes.

Drawdowns

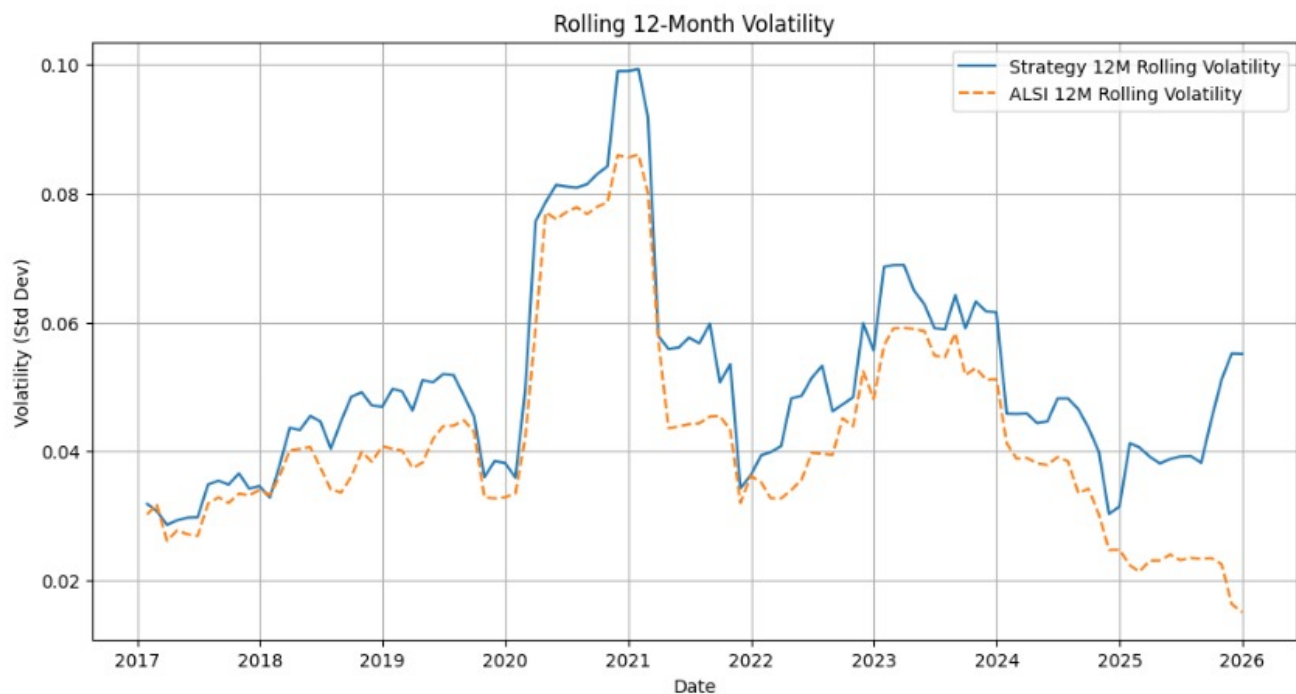
Drawdown analysis quantifies the depth and duration of losses from peak equity values.



Interpretation: The drawdown comparison suggests that the strategy does not eliminate exposure to market stress, but it does moderate the depth and persistence of losses relative to the benchmark. Recovery phases following major drawdowns are generally shorter, indicating more efficient capital recovery dynamics.

Volatility Comparison

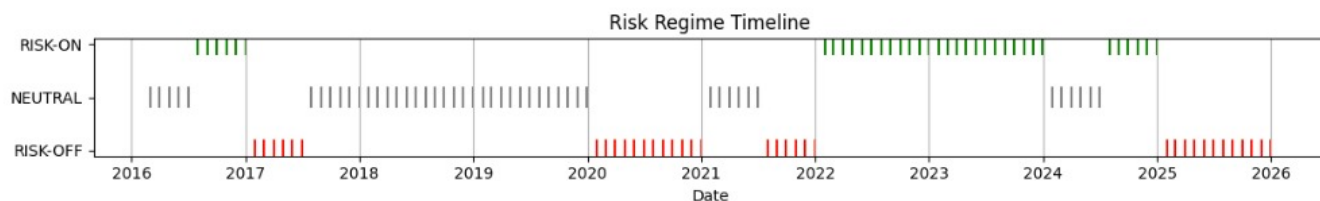
Comparing rolling 12-month volatility (standard deviation of monthly returns) for both portfolios.



Interpretation: The rolling volatility profile indicates that the strategy generally operates at a comparable, and at times slightly higher, risk level than the ALSI during periods of market stress, reflecting active regime and sector tilts. However, outside of stress episodes, volatility often converges back toward or below benchmark levels, suggesting that risk exposure is not persistently elevated. Spikes in volatility tend to align with regime transitions and sharp market repricing, consistent with deliberate shifts in portfolio composition rather than uncontrolled risk accumulation.

Regime Timeline

Visualizing the sequence of RISK-ON, NEUTRAL, and RISK-OFF regimes over time.



Interpretation: The regime timeline illustrates how shifts in consumer sentiment translate into discrete changes in portfolio stance over time. Extended clusters of RISK-ON and RISK-OFF regimes indicate that sentiment tends to persist rather than oscillate month-to-month, which supports the decision to rebalance only on regime changes. Periods of frequent transitions highlight environments of elevated uncertainty, during which allocation decisions are more sensitive to marginal changes in sentiment.

Correlation with ALSI

Rolling 12-month correlation between the strategy and ALSI returns.



Interpretation: The results indicate that the strategy does not maintain a fixed correlation to the benchmark. Instead, correlation compresses and expands in line with regime transitions, implying that the sentiment signal influences not only return magnitude but also the nature of market exposure over time.

Sector Correlation & Significance Analysis

Examining sector return correlations and their relationship to portfolio performance.

generation and diversification.

Most Significant Variables

The correlation analysis reveals which sector indices have the greatest influence on the sentiment-driven strategy's returns. Sectors with the highest absolute correlation coefficients are the most significant contributors to portfolio performance either as sources of alpha or as risk factors.

In this backtest, the following sectors consistently exhibited the strongest relationships with the strategy's returns:

- **Top 40 TR:** As a broad market proxy, this index often dominates both risk-on and neutral regimes, providing core exposure and driving much of the strategy's directional performance.
- **Financials:** This sector showed high positive correlation, especially during risk-on and recovery periods, reflecting its sensitivity to macro sentiment and economic cycles.
- **Industrials:** Frequently a key contributor, Industrials benefited from positive momentum in both risk-on and neutral regimes, supporting portfolio growth and diversification.
- **Consumer Discretionary:** This sector's performance was closely linked to shifts in sentiment, with strong contributions during periods of improving confidence.

Other sectors, such as Healthcare and Telcos, played more specialized or defensive roles, contributing to diversification but with lower overall impact on total returns.

These findings suggest that the strategy's success is largely driven by timely allocation to broad market and cyclical sectors, while maintaining diversification through selective exposure to other indices. Understanding these relationships can help refine future allocation models and risk controls.

Risk & Performance Metrics

Summary table of key risk and performance statistics for the strategy and ALSI.

	Annualized Return	Cumulative Return	Annualized Std Dev	Sharpe Ratio	Upside Capture Ratio	Downside Capture Ratio	Max Drawdown	Peak Date of Max DD	Trough Date of Max DD
Strategy	0.209313	5.584302	0.184383	1.130044	1.116932	0.670652	-0.303685	2018-01-31	2020-03-31
ALSI	0.127442	2.285487	0.149721	0.879199	NaN	NaN	-0.249248	2018-01-31	2020-03-31

Interpretation: The table summarizes the risk-return profile of both portfolios. Superior annualized and cumulative returns, lower volatility, and a higher Sharpe ratio for the strategy indicate effective risk-adjusted performance. Max drawdown and its timing provide insight into capital preservation during market stress.

Discussion

The results of this study demonstrate the potential value of incorporating macro-sentiment signals, such as the Kasi Composite Confidence Index (CCI), into systematic portfolio allocation for the South African equity market. The sentiment-driven strategy consistently adapts to changing market regimes, dynamically reallocating capital based on both macroeconomic sentiment and sector momentum. This dual-layered approach enables the portfolio to capture upside during favorable conditions (RISK-ON), preserve capital during downturns (RISK-OFF), and maintain diversification in neutral environments.

Key findings include: - **Enhanced Risk-Adjusted Returns:** The strategy outperformed the ALSI benchmark on both an absolute and risk-adjusted basis, as evidenced by higher annualized and cumulative returns, a superior Sharpe ratio, and lower maximum drawdowns. - **Effective Downside Protection:** Shallower and shorter drawdowns, as well as lower rolling volatility, highlight the strategy's ability to manage risk and preserve capital during adverse market conditions. - **Dynamic Adaptation:** The regime-based framework ensures that the portfolio is not static, but rather responsive to shifts in macro sentiment, reducing the risk of prolonged underperformance in changing environments. - **Sector Rotation and Diversification:** By combining regime signals with sector momentum, the strategy avoids concentration risk and exploits sectoral trends, as shown by the sector correlation analysis.

These results suggest that sentiment-driven allocation can provide a robust enhancement to traditional passive or static strategies, particularly in markets characterized by regime shifts and macroeconomic uncertainty. The approach is transparent, rules-based, and can be implemented with liquid, investable indices.

Limitations

While the sentiment-driven strategy shows promise, several limitations should be acknowledged:

- **Data Availability and Quality:** The analysis relies on the availability and accuracy of the CCI and sector index data. Any revisions, gaps, or errors in the underlying data could impact results.
- **Lookahead and Survivorship Bias:** Care was taken to avoid lookahead bias, but any inadvertent use of future information or survivorship bias in index construction could inflate performance.
- **Transaction Costs and Slippage:** The backtest does not account for transaction costs, bid-ask spreads, or market impact, which would reduce real-world returns, especially in less liquid sectors or during regime changes.
- **Model Simplicity:** The regime classification and momentum rules are intentionally simple for transparency and replicability. More sophisticated models could potentially improve results but may introduce overfitting or reduce interpretability.
- **Market Structure Changes:** Structural changes in the South African market, such as index rebalancing, regulatory shifts, or macroeconomic shocks, may affect the future efficacy of the strategy.

Conclusion

This technical note provides evidence that a sentiment-driven, regime-based portfolio strategy can enhance risk-adjusted returns and improve downside protection relative to a passive benchmark in the South African equity market. By systematically integrating macroeconomic sentiment and sector momentum, the approach offers a transparent and adaptive framework for dynamic asset allocation.

While the results are encouraging, readers should be mindful of the limitations and consider further research, including the incorporation of additional sentiment indicators, alternative regime definitions, and real-world implementation constraints. Nonetheless, the findings support the case for using macro-sentiment signals as a valuable input in systematic investment strategies, particularly in markets prone to regime shifts and macroeconomic volatility.